

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
    COUNT(DISTINCT(customer_id)) AS total_customers
FROM order_t;
```

```
SELECT
    state,
    COUNT(customer_id) AS total_customers
FROM customer_t
GROUP BY state
ORDER BY total_customers DESC;
```

Output:

state	total_customers
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35

Observations and Insights:

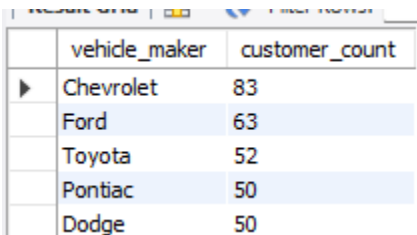
- There's a total of 994 customers, 35% of the total live in California, Texas, Florida, and New York.
- The data appears to be in a uniform distribution, with a 49% drop off in customers from the 4th state with the most customers to the 5th state, but the remaining states have similar customer counts.
- The data is likely showing that population is a driving factor to the number of customers on the new wheels app, since the top 4 states with the most customers happen to be the 4 most populated states.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT  
  
    vehicle_maker,  
  
    COUNT(customer_t.customer_id) AS customer_count  
  
FROM product_t  
  
JOIN order_t  
  
ON product_t.product_id = order_t.product_id  
  
JOIN customer_t  
  
ON order_t.customer_id = customer_t.customer_id  
  
GROUP BY vehicle_maker  
  
ORDER BY customer_count DESC  
  
LIMIT 5;
```

Output:



	vehicle_maker	customer_count
▶	Chevrolet	83
	Ford	63
	Toyota	52
	Pontiac	50
	Dodge	50

Observations and Insights:

- The top 5 vehicle makers are Chevrolet, Ford, Toyota, Dodge, and Pontiac; with Chevrolet being a clear favorite
- Chevrolet is the clear favorite amongst customers, but the distribution is uniform
- Dodge and Pontiac have the same amount of purchases

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT *
FROM
(
SELECT
state,
vehicle_maker,
COUNT(customer_id) AS total_customers,
RANK() OVER (PARTITION BY state ORDER BY COUNT(customer_id) DESC) AS vehicle_ranking
FROM product_t
JOIN order_t USING (product_id)
JOIN customer_t USING (customer_id)
GROUP BY state, vehicle_maker
)
AS preferred_vehicle
WHERE vehicle_ranking = 1
ORDER BY total_customers DESC;
```

Output:

	state	vehide_maker	total_customers	vehide_ranking
▶	Texas	Chevrolet	9	1
	Florida	Toyota	7	1
	California	Ford	6	1
	California	Dodge	6	1
	California	Audi	6	1
	California	Nissan	6	1
	California	Chevrolet	6	1
	Ohio	Chevrolet	6	1
	Alabama	Dodoe	5	1

Observations and Insights:

- California has a 5 way tie for most popular vehicle maker; Nissan, Ford, Dodge, Chevrolet, Audi

- Chevrolet is ranked first in Ohio, California, and Texas, supporting the claim from the previous question that Chevrolet is the most popular brand
- Due to the heavy concentration of customers in California, New Wheels should look to expand marketing pushes within California



Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
WITH customer_rating AS (  
  
SELECT  
  
customer_feedback,  
  
quarter_number,  
  
CASE WHEN customer_feedback = 'very bad' THEN '1'  
  
WHEN customer_feedback = 'bad' THEN '2'  
  
WHEN customer_feedback = 'okay' THEN '3'  
  
WHEN customer_feedback = 'good' THEN '4'  
  
WHEN customer_feedback = 'very good' THEN '5'  
  
END AS total_rating  
  
FROM order_t  
  
)  
  
SELECT  
  
ROUND(AVG(total_rating), 2) AS average_total_rating  
  
FROM customer_rating;  
  
WITH customer_rating AS (  
  
SELECT  
  
customer_feedback,  
  
quarter_number,  
  
CASE WHEN customer_feedback = 'very bad' THEN '1'  
  
WHEN customer_feedback = 'bad' THEN '2'  
  
WHEN customer_feedback = 'okay' THEN '3'
```

```
WHEN customer_feedback = 'good' THEN '4'
WHEN customer_feedback = 'very good' THEN '5'

END AS total_rating

FROM order_t

)

SELECT

quarter_number,

ROUND(AVG(total_rating), 2) AS rating_per_quarter

FROM customer_rating

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

	quarter_number	rating_per_quarter
▶	1	3.55
	2	3.35
	3	2.96
	4	2.4

Observations and Insights:

- The average total rating is estimated 3.14, meaning that the average customer feedback is okay
- Progressively, the customer feedback gets worse, starting from 3.55 in quarter 1, to 2.4 in quarter 4, with a steady decline from quarter to quarter
- This decline indicates customer dissatisfaction overtime

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
WITH feedback_customer AS (  
  
SELECT  
  
quarter_number,  
  
SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) AS VERY_GOOD,  
SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) AS GOOD,  
SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) AS OKAY,  
SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) AS BAD,  
SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) AS VERY_BAD,  
COUNT(customer_feedback) AS FEEDBACK  
  
FROM order_t  
  
GROUP BY quarter_number  
ORDER BY quarter_number  
  
)  
  
SELECT  
  
quarter_number,  
  
ROUND((VERY_GOOD/FEEDBACK), 2) AS VERY_GOOD,  
ROUND((GOOD/FEEDBACK), 2) AS GOOD,  
ROUND((OKAY/FEEDBACK), 2) AS OKAY,  
ROUND((BAD/FEEDBACK), 2) AS BAD,  
ROUND((VERY_BAD/FEEDBACK), 2) AS VERY_BAD  
  
FROM feedback_customer  
  
GROUP BY quarter_number  
ORDER BY quarter_number;
```

Output:

	quarter_number	VERY_GOOD	GOOD	OKAY	BAD	VERY_BAD
▶	1	0.30	0.29	0.19	0.11	0.11
	2	0.29	0.22	0.20	0.14	0.15
	3	0.17	0.21	0.22	0.23	0.18
	4	0.10	0.10	0.20	0.29	0.31

Observations and Insights:

- 'Very good' is the highest feedback for the first 2 quarters, 'okay' is the highest for Q3 and 'very bad' is the highest for Q4
- "bad" and 'very bad' were tied for the lowest rated feedback for quarter 1-2, and 'very good' became the lowest rated the last two quarters
- Shows a complete flip in customer satisfaction as the year goes on

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
    COUNT(order_id) AS TOTAL_ORDERS  
  
FROM order_t  
  
GROUP BY quarter_number  
  
ORDER BY quarter_number;
```

Output:

	quarter_number	TOTAL_ORDERS
▶	1	310
	2	262
	3	229
	4	199

Observations and Insights:

- The first quarter yielded the highest total orders than any other quarter
- Following the first quarter, the total number of orders declined at a steady rate
- The decrease in sales is likely due to gradual decrease in customer satisfaction, as founded in the previous questions.

Question 7: Calculate the net revenue generated by the company.
What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
WITH quart_rev AS (

    SELECT

        quarter_number,

        ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * vehicle_price))), 0) AS revenue

    FROM

        order_t

    GROUP BY

        quarter_number

)

SELECT

    quarter_number,

    revenue,

    LAG(revenue) OVER (ORDER BY quarter_number) AS previous_revenue,

    ROUND(

        (revenue - LAG(revenue) OVER (ORDER BY quarter_number)) /

        LAG(revenue) OVER (ORDER BY quarter_number),

        2) AS quart_perc_change

    FROM

        quart_rev;
```

Output:

	quarter_number	revenue	previous_revenue	quart_perc_change
▶	1	39421580	NULL	NULL
	2	32715830	39421580	-0.17
	3	29229896	32715830	-0.11
	4	23346780	29229896	-0.20

Observations and Insights:

- First quarter has the highest net revenue of any other quarter, with a steady decline after that
- Similarly, there is a quarter over quarter percent change in each quarter, showing that net revenue is decreasing each quarter
- The largest drop off in revenue appears to be Q2 to Q3, which is around the same time that customers expressed dissatisfaction in their feedback
- Likewise, the continued drop off in revenue directly correlates with the previous findings in a decrease in orders from the beginning of the year to the end.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
    COUNT(order_id) AS total_orders,  
  
    SUM(vehicle_price * quantity) AS total_revenue  
  
FROM order_t  
  
GROUP BY quarter_number  
  
ORDER BY quarter_number ASC;
```

Output:

	quarter_number	total_orders	total_revenue
▶	1	310	39637630.97
	2	262	32913737.76
	3	229	29435427.48
	4	199	23496008.22

Observations and Insights:

- Similarly to what we saw with net revenue, total orders peaks with Q1 and decreased from Q2 to Q4
- The biggest drop off in orders appears to be from Q1 to Q2, a

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    credit_card_type,
    ROUND(AVG(discount), 2) AS average_discount
FROM order_t t1
INNER JOIN customer_t t2
ON t1.customer_id = t2.customer_id
GROUP BY credit_card_type
ORDER BY average_discount DESC;
```

Output:

	credit_card_type	average_discount
▶	laser	0.64
	mastercard	0.63
	visa-electron	0.62
	china-unionpay	0.62
	americanexpress	0.62
	maestro	0.62
	instapayment	0.62
	...	...

Observations and Insights:

- The highest average discount is laser with 0.64%
- Visa-electron, china-unionpay, americanexpress, maestro, and instapayment share the same value of 0.62.
- The discounts appear very uniformly distributed

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
    ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS average_shipping_time  
  
FROM order_t  
  
GROUP BY quarter_number  
  
ORDER BY AVERAGE_SHIPPING_TIME;
```

Output:

	quarter_number	average_shipping_time
▶	1	57.17
	2	71.11
	3	117.76
	4	174.10

Observations and Insights:

- Q1 is the most efficient quarter in terms of shipping day
- From Q1 to Q4, shipping time increases drastically
- This is interesting because the past statistics revealed spike in orders during Q1 and decline Q2 to Q4.
- It would be likely to assume that the decrease in customer satisfaction is due to the rise in shipping time

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
< '125482804.43'>	<1000>	<994>	<3.07>
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
<23496008.22>	< 199>	<105.04>	<20%>

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- Improve overall employee performance, start by getting the shipping time down. This would result in greater customer satisfaction and positive word of mouth, leading to more new customers.
- Offer promotions to new and existing customers to get the order numbers back up and create positive word of mouth for the business
- Reach out to customers from Q1 for more feedback or testimonials to help restore faith in New Wheels among the fading customer base.